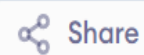


# Medium – Matrix Operations

main.c



Run

```
1  #include <stdio.h>
2
3  #define MAX 10
4
5  // Function to input a matrix
6  void inputMatrix(int matrix[MAX][MAX], int rows, int cols) {
7      int i, j;
8      printf("Enter matrix elements:\n");
9      for(i = 0; i < rows; i++) {
10         for(j = 0; j < cols; j++) {
11             scanf("%d", &matrix[i][j]);
12         }
13     }
14 }
15
16 // Function to display a matrix
17 void displayMatrix(int matrix[MAX][MAX], int rows, int cols) {
18     int i, j;
19     for(i = 0; i < rows; i++) {
20         for(j = 0; j < cols; j++) {
21             printf("%d\t", matrix[i][j]);
22         }
23         printf("\n");
24     }
```

```

15
16 // Function to display a matrix
17 void displayMatrix(int matrix[MAX][MAX], int rows, int cols) {
18     int i, j;
19     for(i = 0; i < rows; i++) {
20         for(j = 0; j < cols; j++) {
21             printf("%d\t", matrix[i][j]);
22         }
23         printf("\n");
24     }
25 }
26
27 // Matrix Addition
28 void addMatrices(int A[MAX][MAX], int B[MAX][MAX], int result[MAX][MAX],
29     int rows, int cols) {
30     int i, j;
31     for(i = 0; i < rows; i++) {
32         for(j = 0; j < cols; j++) {
33             result[i][j] = A[i][j] + B[i][j];
34         }
35     }
36

```

```

35 }
36
37 // Matrix Multiplication
38 void multiplyMatrices(int A[MAX][MAX], int B[MAX][MAX], int
39     result[MAX][MAX], int r1, int c1, int c2) {
40     int i, j, k;
41     for(i = 0; i < r1; i++) {
42         for(j = 0; j < c2; j++) {
43             result[i][j] = 0;
44             for(k = 0; k < c1; k++) {
45                 result[i][j] += A[i][k] * B[k][j];
46             }
47         }
48     }
49 }
50
51 // Matrix Transpose
52 void transposeMatrix(int matrix[MAX][MAX], int transpose[MAX][MAX], int
53     rows, int cols) {
54     int i, j;
55     for(i = 0; i < rows; i++) {
56         for(j = 0; j < cols; j++) {

```

```

51 // Matrix Transpose
52 void transposeMatrix(int matrix[MAX][MAX], int transpose[MAX][MAX], int
    rows, int cols) {
53     int i, j;
54
55     for(i = 0; i < rows; i++) {
56         for(j = 0; j < cols; j++) {
57             transpose[j][i] = matrix[i][j];
58         }
59     }
60 }
61
62 int main() {
63     int A[MAX][MAX], B[MAX][MAX];
64     int sum[MAX][MAX], product[MAX][MAX], transpose[MAX][MAX];
65     int r1, c1, r2, c2;
66
67     // Input dimensions for Matrix A
68     printf("Enter rows and columns of Matrix A: ");
69     scanf("%d %d", &r1, &c1);
70
71     inputMatrix(A, r1, c1);
72

```

```

72
73     // Input dimensions for Matrix B
74     printf("Enter rows and columns of Matrix B: ");
75     scanf("%d %d", &r2, &c2);
76
77     inputMatrix(B, r2, c2);
78
79     // Matrix Addition
80     if(r1 == r2 && c1 == c2) {
81         addMatrices(A, B, sum, r1, c1);
82
83         printf("\nMatrix Addition Result:\n");
84         displayMatrix(sum, r1, c1);
85     } else {
86         printf("\nMatrix Addition not possible!\n");
87     }
88
89     // Matrix Multiplication
90     if(c1 == r2) {
91         multiplyMatrices(A, B, product, r1, c1, c2);
92
93         printf("\nMatrix Multiplication Result:\n");
94         displayMatrix(product, r1, c2);

```

```

84         displayMatrix(sum, r1, c1);
85     } else {
86         printf("\nMatrix Addition not possible!\n");
87     }
88
89     // Matrix Multiplication
90     if(c1 == r2) {
91         multiplyMatrices(A, B, product, r1, c1, c2);
92
93         printf("\nMatrix Multiplication Result:\n");
94         displayMatrix(product, r1, c2);
95     } else {
96         printf("\nMatrix Multiplication not possible!\n");
97     }
98
99     // Transpose of Matrix A
100    transposeMatrix(A, transpose, r1, c1);
101
102    printf("\nTranspose of Matrix A:\n");
103    displayMatrix(transpose, c1, r1);
104
105    return 0;
106 }

```

## RESULT...

Enter rows and columns of Matrix A: 2 2

Enter matrix elements:

23 43

12 23

Enter rows and columns of Matrix B: 2 2

Enter matrix elements:

1 2

5 6

Matrix Addition Result:

24 45

17 29

Matrix Multiplication Result:

238 304

127 162

Transpose of Matrix A:

23 12

43 23

Enter rows and columns of Matrix A: 3 3

Enter matrix elements:

12 12 12

3 2 2

45 2 23

Enter rows and columns of Matrix B: 3 3

Enter matrix elements:

12 25 13

12

44

2

12 1 12

Matrix Addition Result:

24 37 25

15 46 4

57 3 35

Matrix Multiplication Result:

432 840 324

84 165 67

840 1236 865